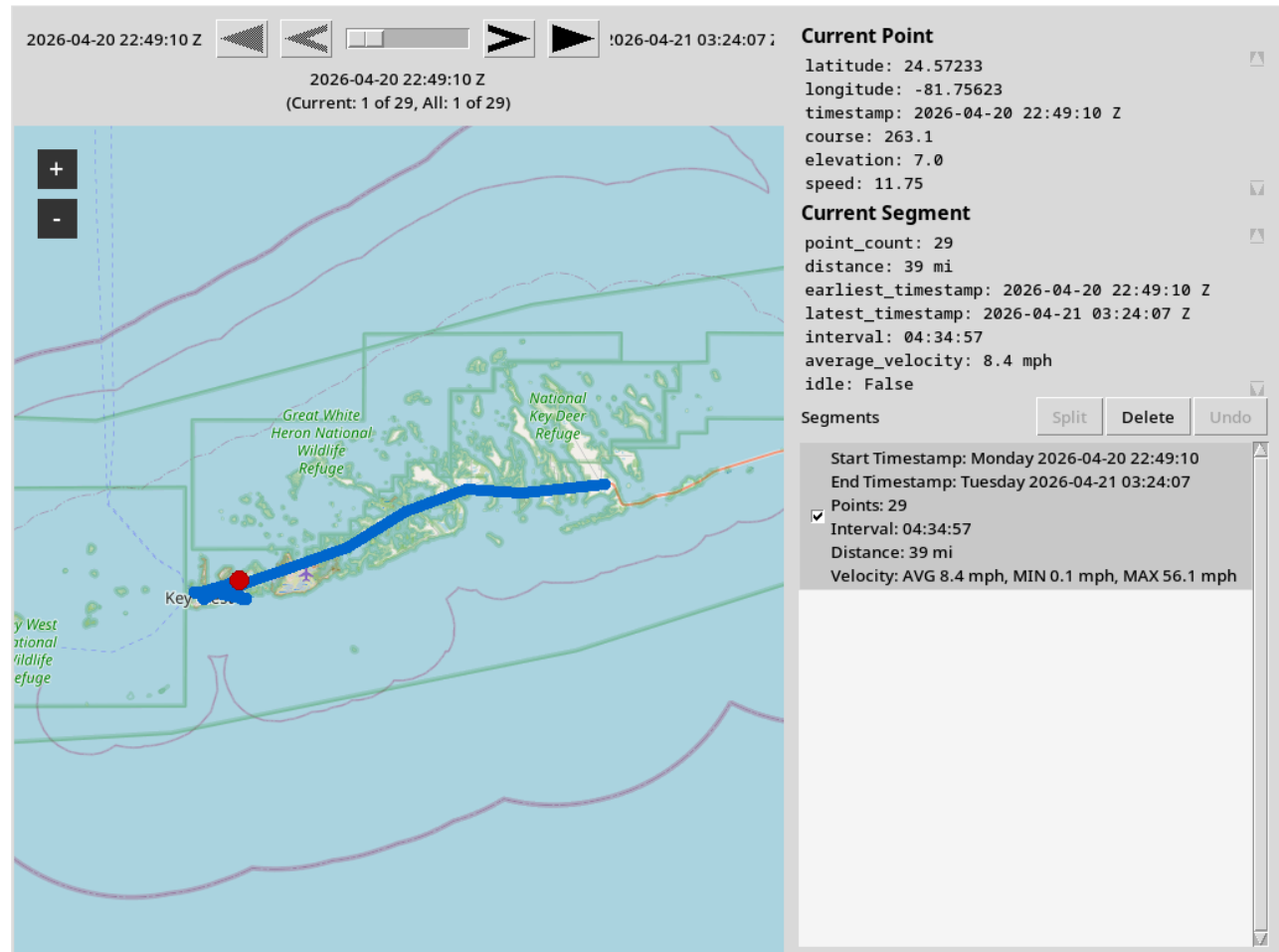


Features

Complete catalog of GIS Graphical Editor (gge) capabilities, including map display, time-slider animation, segment splitting and deletion, metadata panels, and optional point and interval overlays.



Application shell

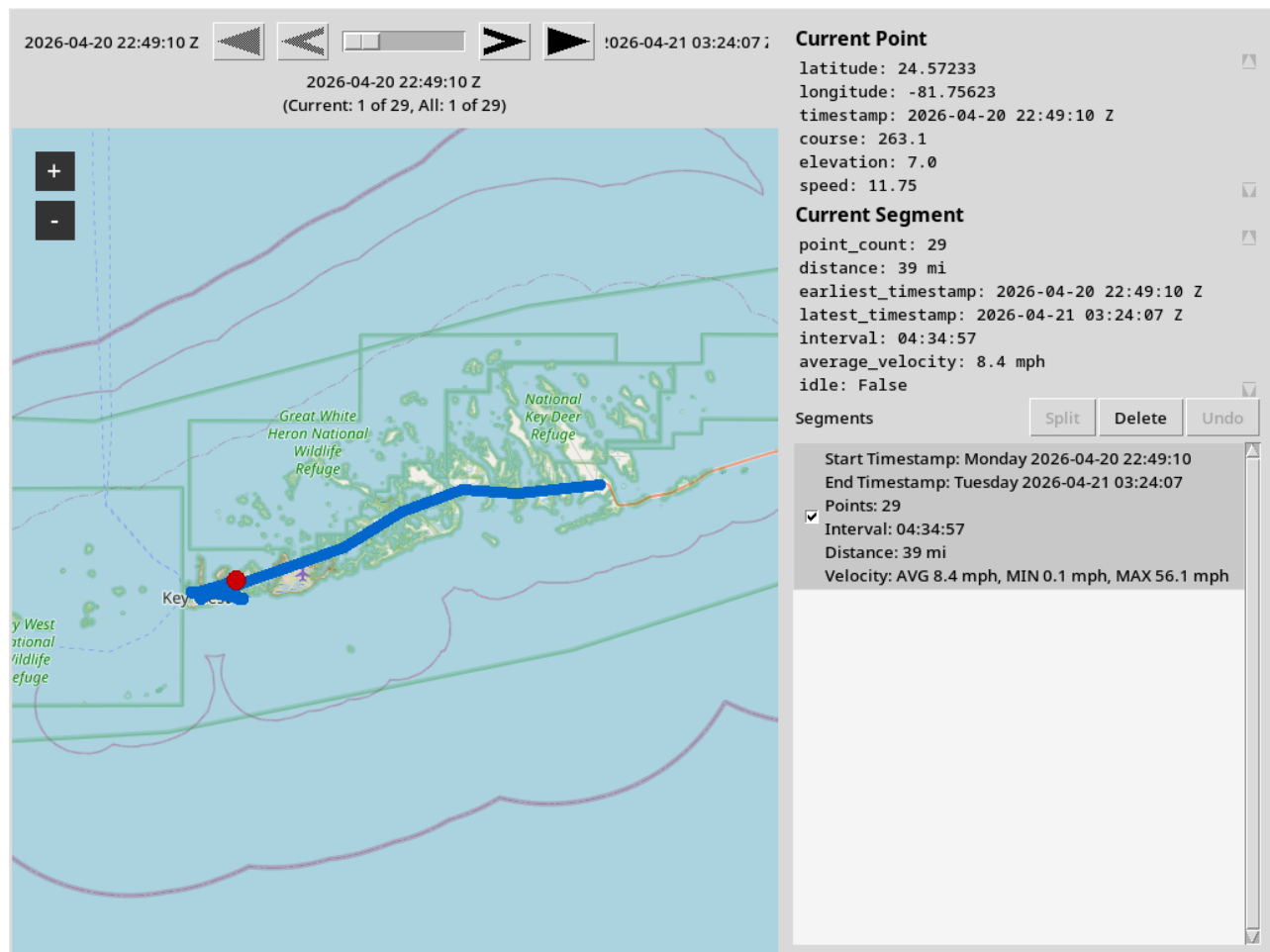
- **Desktop GUI** built with Tkinter and [tkintermapview](#).
- **Default window** size 1024×768 with title "GIS Graphical Editor".
- **Console entry point** gge (package name gge, import path gis_graphical_editor).
- **Lazy map creation:** the map widget is created only after a GPX file is successfully loaded; closing the track destroys the widget.

GPX loading

- **File dialog load** via **File → Load** (Ctrl+O) or startup prompt when no --filepath is given.

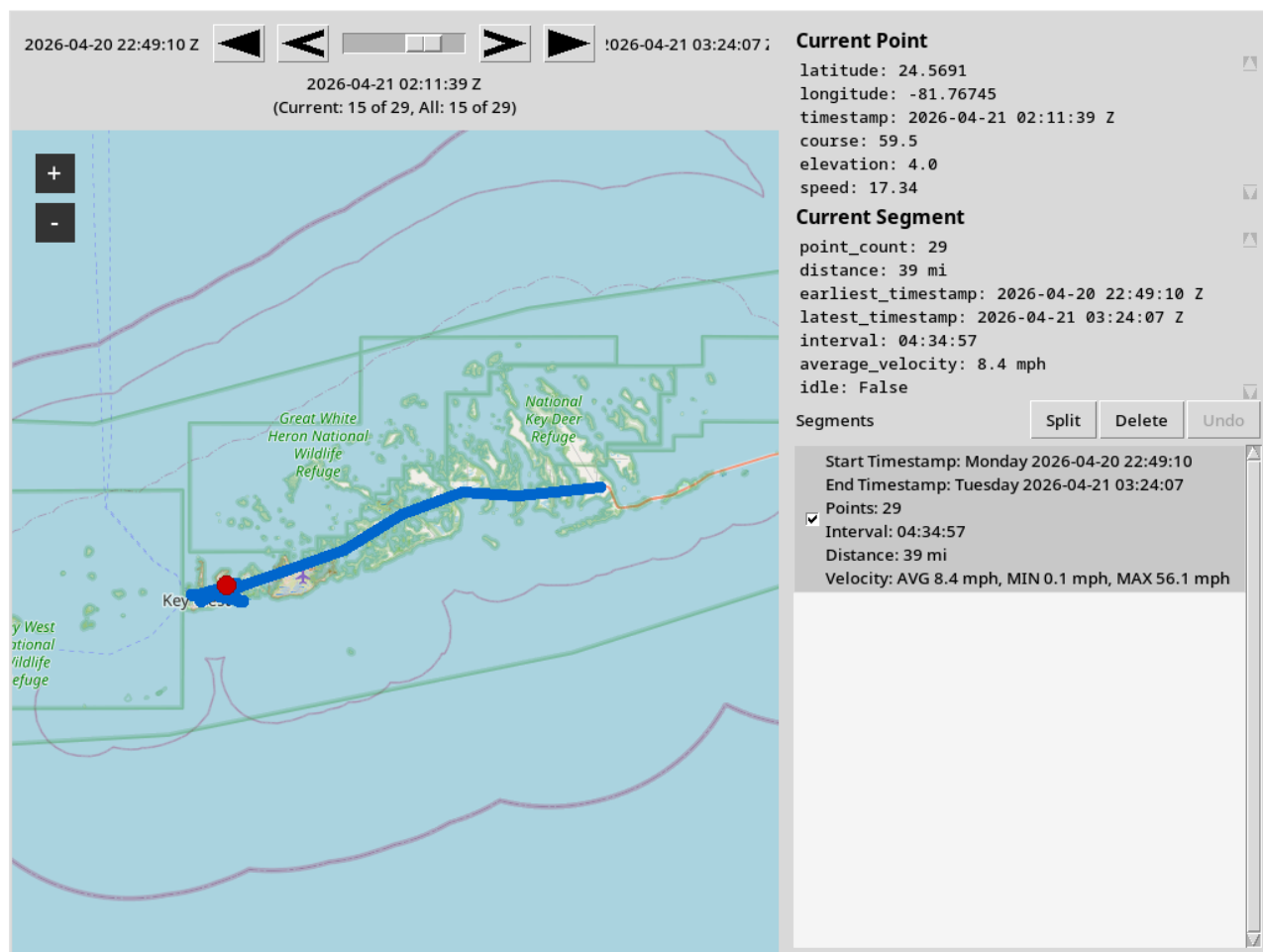
- **Startup load** via `--filepath PATH` without opening the dialog first.
- **Supported geometry sources** (first non-empty wins):
 - Track segment points, in file order across all tracks and segments.
 - Route points when the file has no track segments.
 - Standalone waypoints when tracks and routes are absent.
- **Point model** (`GpxPointRecord`): latitude, longitude, and optional timestamp per point.
- **Timestamp normalization**: timezone-aware GPX timestamps are converted to naive datetimes for analysis.
- **--as-timezone IANA_NAME**: convert all loaded timestamps to the named IANA zone for display; warns when naive timestamps are interpreted as UTC first.
- **Error feedback**:
 - Error dialog when the path is not a file.
 - Error dialog when the GPX cannot be parsed.
 - Warning dialog when no track points are found.
 - Warning when `--mark-hours` is set but the loaded GPX has no timestamps (hour markers are skipped; the path still draws).
 - Warning when the GPX has no timestamps (the time slider, metadata panel, and segment list are not shown).

Map display



- **Base map:** OpenStreetMap tiles through tkintermapview.
- **Track path:** blue polyline (#0066CC, width 9) through all loaded coordinates.
- **Auto framing:** after each load, the map fits a bounding box around the track (max/min latitude and longitude).
- **Double-click zoom:** double-clicking the map canvas increases zoom by one level, centered on the click position.
- **Close track:** **File** → **Close** (Ctrl+W) clears paths, markers, and removes the map widget. The menu item is disabled until a track is loaded.

Time slider and track animation



- **Requires timestamps** on the loaded GPX; omitted when the file has no timed points.
- **Horizontal slider** above the map spanning earliest to latest track timestamp, with endpoint labels showing date, time, and timezone when available.
- **Current position label** under the slider shows the selected timestamp plus point index within the current segment and across all visible timed points, e.g. (Current: 1 of 29, All: 1 of 29).
- **VCR-style controls:**
 - **Rewind play** — auto-step backward through timed points every 250 ms until the first point; click again to stop.
 - **Previous point** — move to the prior timed GPX point.
 - **Next point** — move to the next timed GPX point.
 - **Fast-forward play** — auto-step forward through timed points every 250 ms until the last point; click again to stop.
 - **Manual scrubbing** stops any active auto-play.
 - **Slider pointer marker:** large red dot on the map at the interpolated position for the selected time.

- **Map follow:** when the pointer would fall outside the visible map tiles, the view pans to keep it on screen (with a small margin).
- **Segment highlight sync:** the segment checklist row containing the current slider time is highlighted in gray.

Track metadata panel

7: **Current Point**

latitude: 24.5691
longitude: -81.76745
timestamp: 2026-04-21 02:11:39 Z
course: 59.5
elevation: 4.0
speed: 17.34

Current Segment

point_count: 29
distance: 39 mi
earliest_timestamp: 2026-04-20 22:49:10 Z
latest_timestamp: 2026-04-21 03:24:07 Z
interval: 04:34:57
average_velocity: 8.4 mph
idle: False

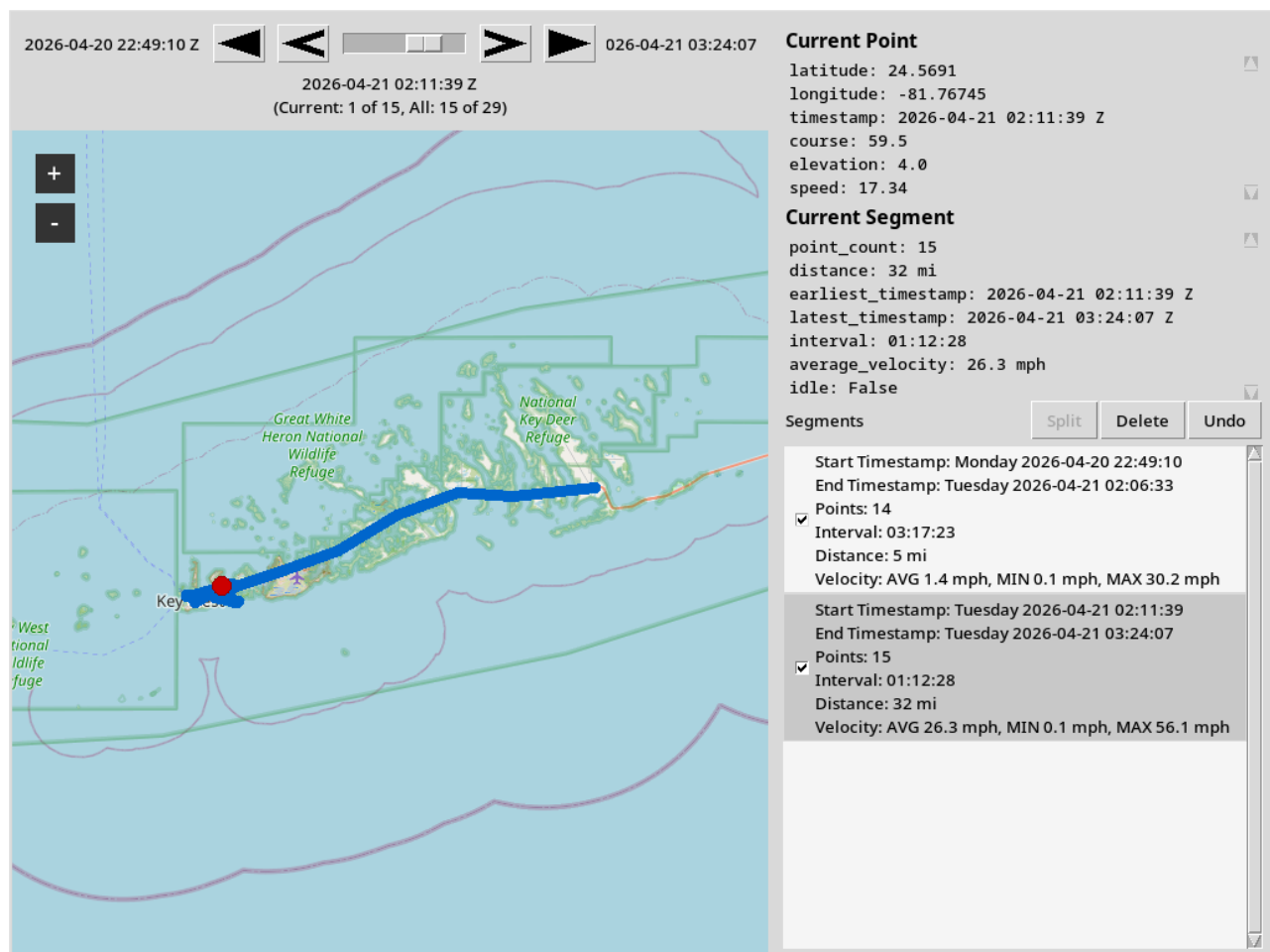
Segments Split Delete Undo

Start Timestamp: Monday 2026-04-20 22:49:10
End Timestamp: Tuesday 2026-04-21 03:24:07
☒ Points: 29
Interval: 04:34:57
Distance: 39 mi
Velocity: AVG 8.4 mph, MIN 0.1 mph, MAX 56.1 mph

- **Requires timestamps;** shown in the right sidebar above the segment list when the time slider is available.

- **Current Point** read-only box lists latitude, longitude, timestamp, and any GPX fields present on the nearest timed point (elevation, speed, course, and similar metadata).
- **Current Segment** read-only box lists point count, path distance, earliest and latest timestamps, elapsed interval, average velocity, and idle status for the segment containing the slider position.
- **Live updates** as the slider moves, segment visibility changes, or segment edits refresh the track.

Segment list and editing



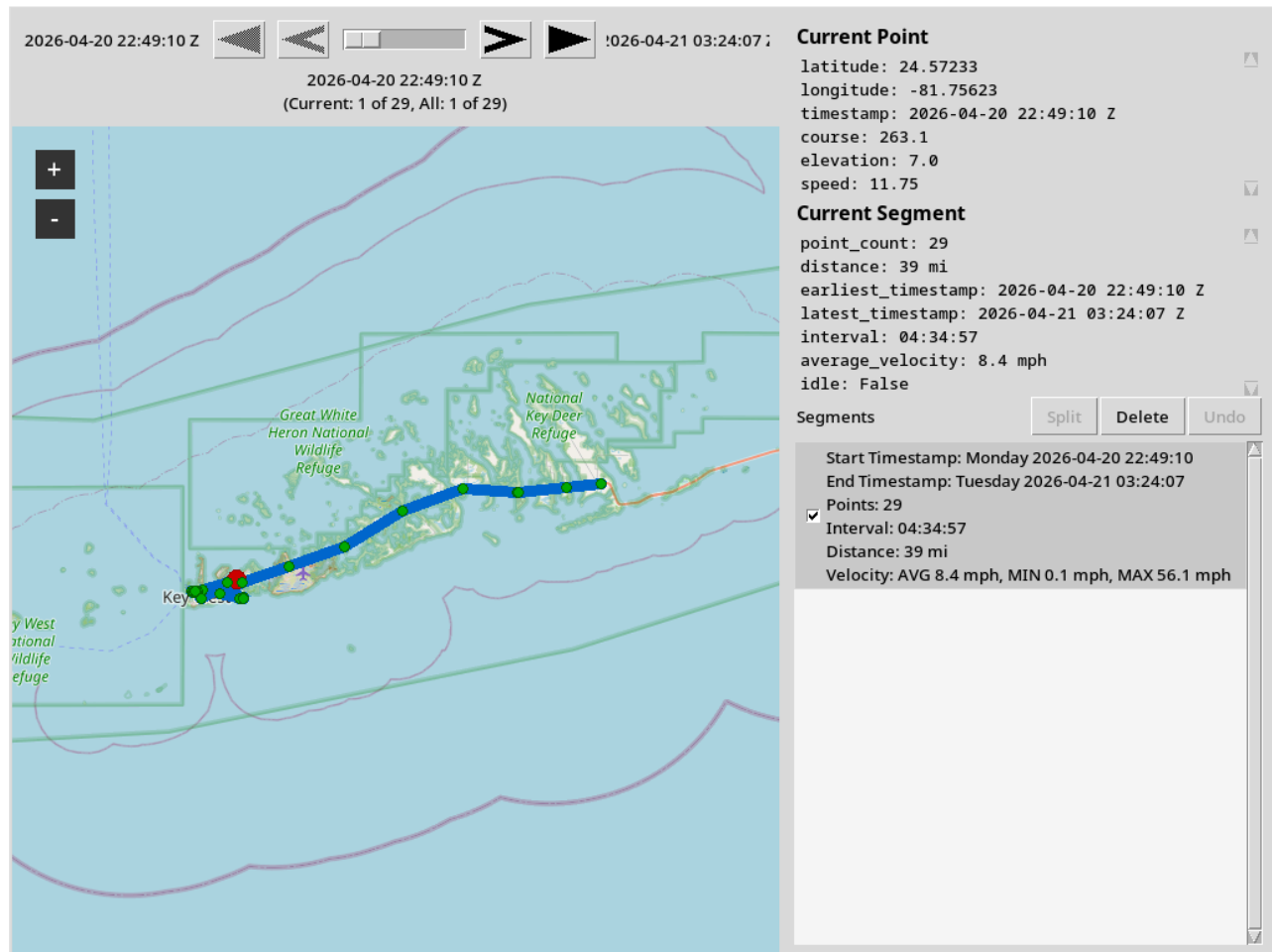
- **Requires timestamps**; scrollable checklist in the right sidebar listing every loaded segment sorted by earliest timestamp.
- **Segment labels** summarize start/end timestamps (with day of week when dated), point count, elapsed interval, path distance, and min/avg/max velocity; idle legs can be omitted from velocity stats when --no-idle is set at launch.
- **Visibility checkboxes** control which segments contribute to the drawn path, overlays, slider, and metadata; unchecked segments remain in memory and in **Save As** export.

- **Split** — divide the highlighted segment at the slider's nearest GPX point into head and tail segments. Disabled at segment endpoints or when no segment context is available.
- **Delete** — remove the highlighted segment after a confirmation dialog describing its timestamp span and point count.
- **Undo** — restore the segment structure and checkbox state from before the most recent split or delete (one level; disabled until an edit has been made).
- `--no-idle` at launch excludes segments whose average velocity is below 1 mph (2 kph when `--metric`) from the checklist entirely.

GPX export

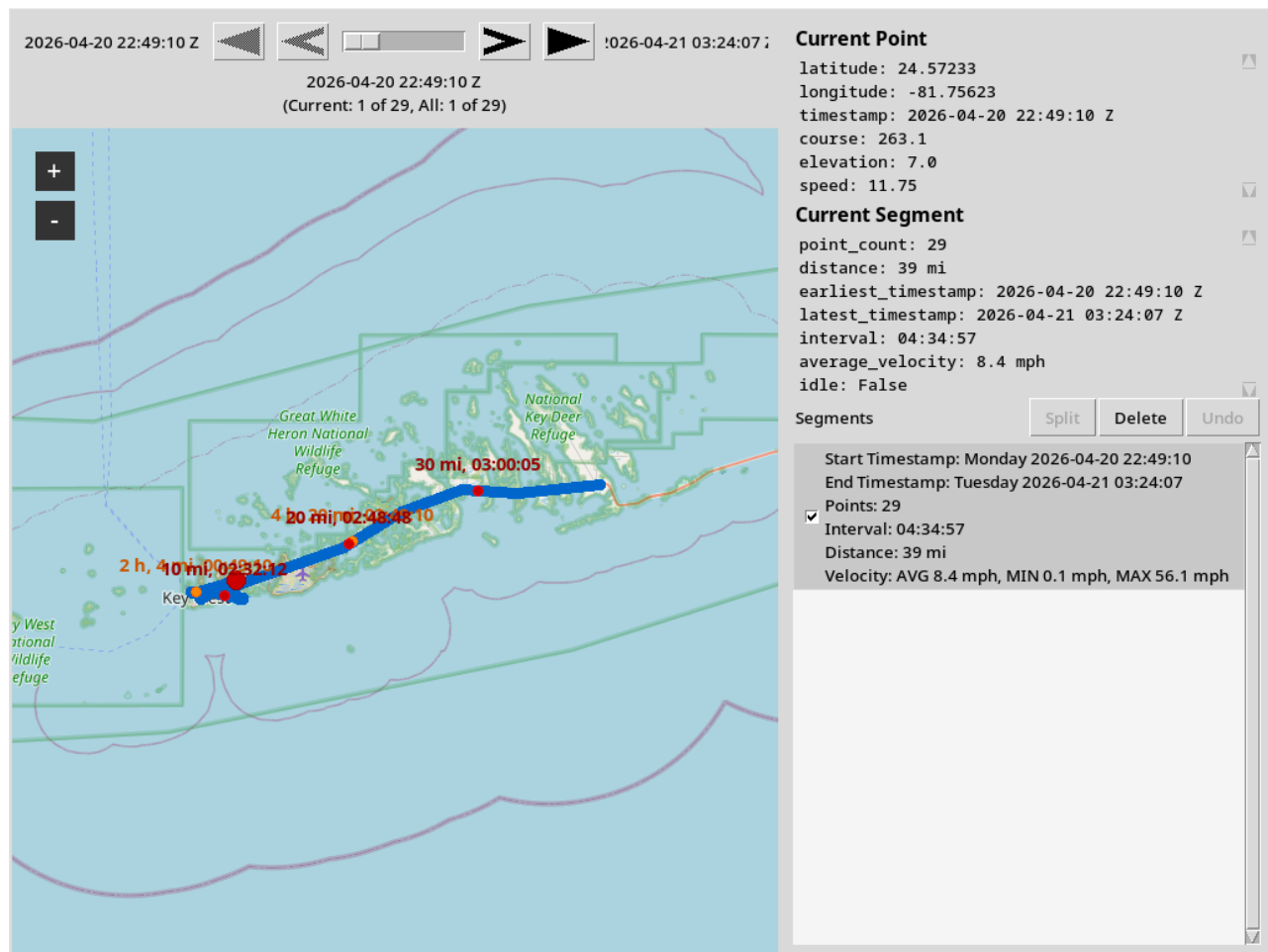
- **Save As** via **File → Save As** (Ctrl+Shift+S) writes the in-memory track to a user-chosen .gpx path.
- Exports **all** segments in `_loaded_gpx_segments`, preserving segment boundaries after split/delete edits.
- Segment checklist checkboxes affect map display only; unchecked segments remain in the exported file.
- Standard GPX point fields (elevation, speed, course, and similar metadata) round-trip through save and reload.
- Error dialog when the file cannot be written.
- Menu item disabled until a track with at least one point is loaded.

Recorded point overlay (--points)



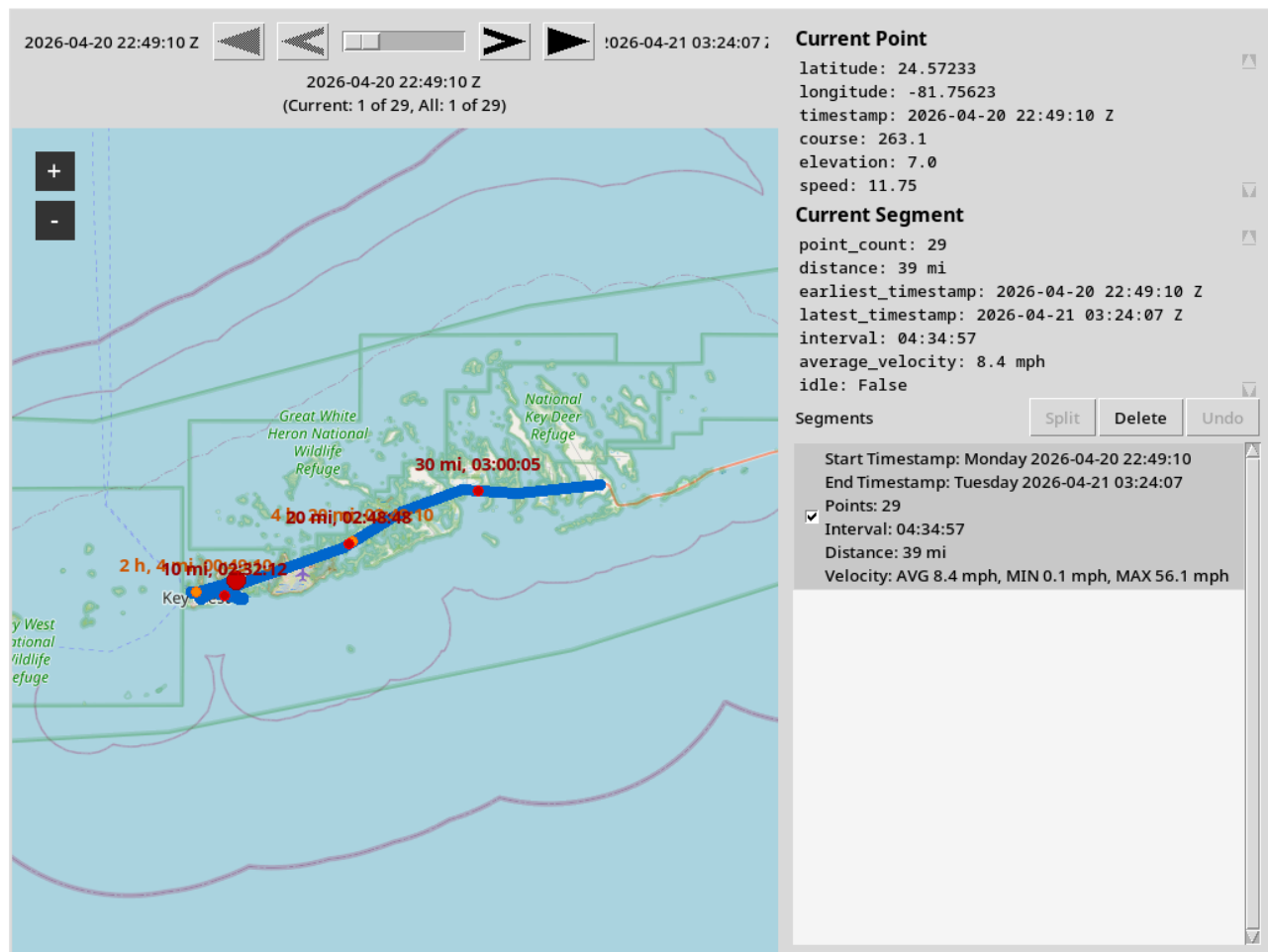
- **Green dot** at every loaded GPX coordinate.
- Icons are slightly wider than the path line for visibility.
- Independent of timestamps (all points are drawn).

Hour interval markers (--mark-hours N)



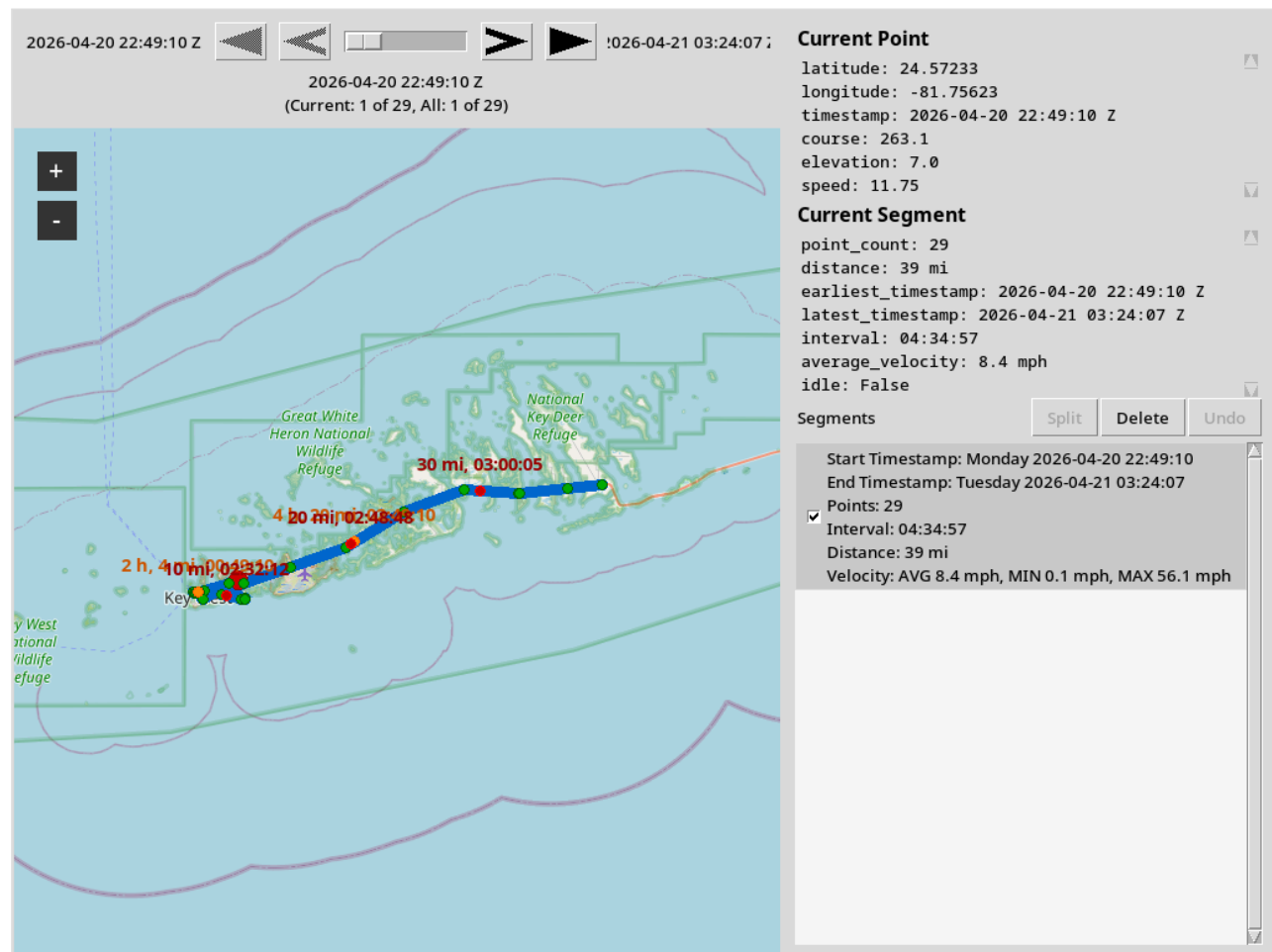
- **Orange dot** placed every N hours along the path (N must be a positive integer).
- **Requires timestamps** on GPX points; segments without both endpoint timestamps are skipped.
- **Position interpolation**: marker latitude/longitude are linearly interpolated along the segment that crosses each hour boundary.
- **Default labels** (unless `--no-mark-labels`): rounded total hours and rounded cumulative distance, e.g. 2 h, 15 mi (kilometers when `--metric`). Include calendar dates when `--dates` is set.
- **Distance along path** for labels uses haversine great-circle distance between consecutive points.

Distance interval markers (--mark-distance N)



- **Red dot** placed every N miles along the path (N must be a positive integer; kilometers when --metric).
- **Does not require timestamps** for placement (distance-only segments are used).
- **Position interpolation:** marker coordinates are linearly interpolated at each mile boundary along a segment.
- **Default labels** (unless --no-mark-labels): rounded total distance and interpolated timestamp when available, e.g. 10 mi, 2024-06-01 09:30:00. Distance-only label when no timestamp can be interpolated. Include calendar dates when --dates is set.
- **Timestamp interpolation:** linear between segment endpoint timestamps when both exist; otherwise falls back to whichever endpoint has a time.

Marker appearance



Marker type	Color	Trigger
Recorded point	Green (#00AA00)	--points
Hour interval	Orange (#FF8800)	--mark-hours N
Distance interval	Red (#CC0000)	--mark-distance N

- All marker icons are filled ellipses with outline, rendered via Pillow and sized to match path width.
- Interval marker text uses orange (#CC5500) or red (#990000) when labels are shown.

Command-line interface

Flag	Effect
--filepath PATH	Load GPX on startup
--points	Show green dots at every point

Flag	Effect
<code>--mark-hours N</code>	Orange markers every <i>N</i> hours
<code>--mark-distance N</code>	Red markers every <i>N</i> miles (kilometers when <code>--metric</code>)
<code>--no-mark-labels</code>	Icons only; no text on interval markers
<code>--as-timezone IANA_NAME</code>	Display timestamps in the named IANA timezone
<code>--dates</code>	Include calendar dates in interval marker labels
<code>--metric</code>	Use kilometers and km/h instead of miles and mph
<code>--no-idle</code>	Omit idle segments from the segment checklist
<code>--help</code>	Print usage and exit

- **Validation:** non-positive `--mark-hours` or `--mark-distance` values produce an `argparse` error and non-zero exit.
- **Combinable options:** e.g. `--points --mark-hours 1 --mark-distance 5` on the same run.
- **Default label behavior:** interval marker labels are shown unless `--no-mark-labels` is passed.

Menus and keyboard shortcuts

Menu	Item	Shortcut	When available
File	Load	Ctrl+O	Always
File	Save As	Ctrl+Shift+S	After a track with points is loaded
File	Close	Ctrl+W	After a track is loaded
File	Exit	Ctrl+Q	Always

- Load opens a file dialog filtered to `*.gpx` with an "All files" option.
- Save As opens a save dialog filtered to `*.gpx` and writes the current in-memory segments.
- Exit calls `root.quit()` to end the main loop.

Track analysis (library)

Reusable logic in `track_analysis.py`:

- `has_timestamps(gpx_points)` — whether any point carries a time.
- `get_timestamp_range(gpx_points)` — earliest and latest timestamps, if any.
- `build_track_segment_summaries(segment_point_lists, ...)` — sorted `TrackSegmentSummary` list for the segment panel.

- `compute_miles_between_points(first, second) / compute_distance_between_points(...)` — haversine distance (miles or kilometers).
- `compute_hours_between_timestamps(first, second)` — elapsed hours between datetimes.
- `build_hour_interval_markers(gpx_points, interval_hours, ...)` — list of `TrackIntervalMarker`.
- `build_distance_interval_markers(gpx_points, interval_miles, ...)` — list of `TrackIntervalMarker`.
- `find_position_at_timestamp(gpx_points, target_timestamp)` — interpolated map position at a time.
- `split_gpx_segment_point_lists_at_point_index(...)` / `remove_gpx_segment_from_segment_point_lists(...)` — segment edit helpers.
- `format_gpx_point_metadata_lines(gpx_point)` / `format_segment_summary_metadata_lines(...)` — metadata panel text.
- Label formatters for hour, distance, and segment interval display.

GPX utilities (library)

- `load_gpx_points_from_gpx(path)` — ordered `GpxPointRecord` list.
- `load_track_points_from_gpx(path)` — ordered (latitude, longitude) tuples.
- `load_track_point_segments_from_gpx(path)` — segment-ordered point lists preserving GPX `<trkseg>` boundaries.
- `write_track_point_segments_to_gpx(path, segment_point_lists)` — write segment lists as one GPX track.
- `convert_gpx_point_timestamps_to_timezone(gpx_points, timezone_name)` — in-place display timezone conversion.

Configuration object

`TrackDisplayOptions` holds display settings passed from the CLI to the main window:

- `mark_hours_interval`
- `mark_distance_interval`
- `show_points`
- `show_mark_labels`
- `initial_gpx_filepath`
- `as_timezone_name`
- `show_dates_in_mark_labels`
- `use_metric_units`
- `exclude_idle_segments`

Dependencies

Package	Role
gpxpy	GPX parsing
tkintermapview	OSM map widget (pulls in Pillow, requests, etc.)
Pillow (transitive)	Marker icon rendering
pytest (optional)	Test runner

Automated tests

Pytest coverage includes:

- GPX track segment loading, export round-trips, and timezone conversion (`test_gpx_utility.py`)
- Hour and distance interval markers, segment summaries, split/delete, and metadata formatters (`test_track_analysis.py`)
- Track display options (`test_track_display_options.py`)
- Time slider play controls and position labels (`test_time_slider_panel.py`)
- CLI validation for non-positive `--mark-hours` (`tests/entrypoint/test_gge.py`)

Out of scope (current version)

- KMZ/KML import (only `.gpx` is supported).
- Multiple simultaneous tracks with separate styling.
- GUI controls for display options (options are CLI-only at launch).